



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- The capstone project is about applying complete data science methodology to explore, analyze, and predict outcomes using real-world data from the SpaceX Falcon 9 launches dataset. The primary goal is to determine the factors that influence the success of Falcon 9 first-stage landings and develop a model to predict the likelihood of future mission success.

- **Summary of methodologies**

The methodologies include –

- 1) Data collection via API
- 2) Data Wrangling
- 3) Exploratory Data Analysis and Data Visualization
- 4) SQL analysis
- 5) Predictive modeling

- **Summary of all results**

The results include

- Graphs showing correlations, SQL query outputs, Interactive map screenshots, Dash dashboard interface, Confusion matrix or accuracy score from predictive models etc.

Introduction

- **Project background and context**

The success of a Falcon 9 launch significantly impacts SpaceX's ability to offer commercial spaceflight at lower costs.

This project uses real-world SpaceX launch data, accessed through the SpaceX REST API to perform a complete data science workflow from data collection to predictive modeling.

Predicting whether the first-stage landing will be successful helps optimize mission planning and cost efficiency.

- **Problems you want to find answers**

- Can we predict the likelihood of a successful booster landing for future missions?
- How do payload mass, orbit type, and launch site affect landing outcomes?
- How can interactive dashboards and maps help visualize findings effectively?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:

The dataset was obtained from the SpaceX REST API, which provides structured JSON data on all SpaceX launches. A GET request is used to retrieve the data in JSON format. The data was then converted into a Pandas DataFrame for analysis.

- Perform data wrangling
 - Checked for missing values, duplicates, and inconsistent data types. Handled missing values, duplicate values and outliers. Renamed and reformatted columns. Normalized the data for visualization and machine learning tasks.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Selected features such as **payload mass**, **orbit**, **launch site** as predictors. Split the data into **training and testing sets** using `train_test_split()`. Implemented multiple classification models like Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). Models were evaluated using accuracy score, confusion matrix, and classification report and selected the best model.

Data Collection – SpaceX API

Describe how data sets were collected.

Python's requests package was used to gather data from the SpaceX REST API.

Structured JSON data about every previous SpaceX launch was made available using the API.

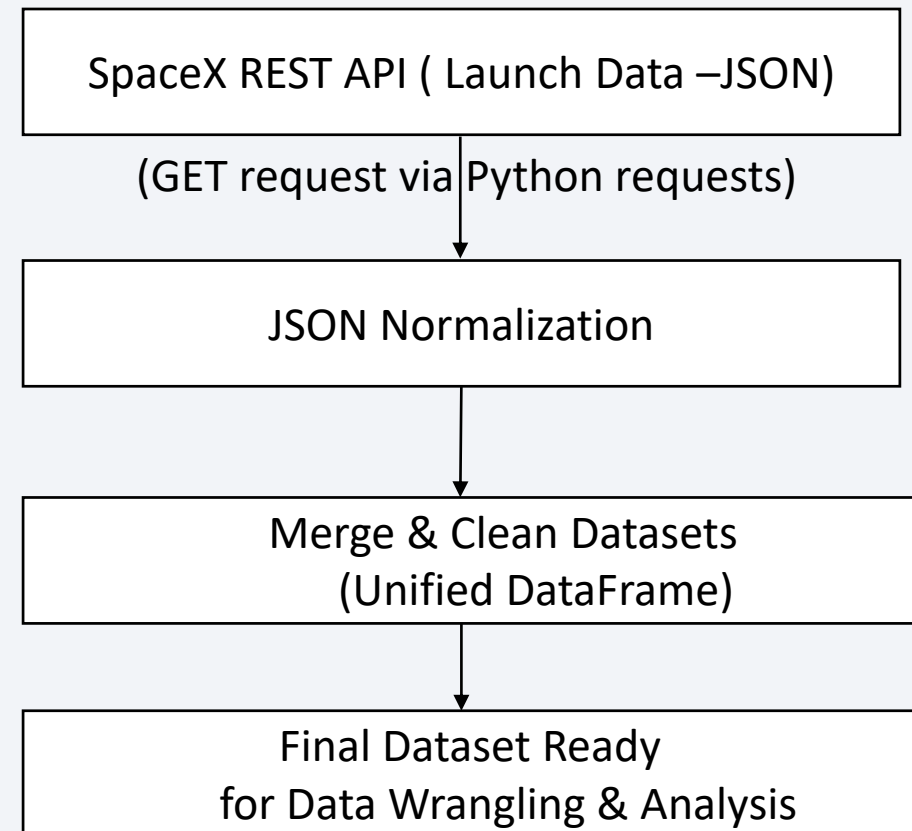
Launch location, rocket type, payload, orbit, landing result, and success status were among the information retrieved.

Pandas was used to normalize the raw JSON response into a tabular format. `json_normalize()`.

Additional information (mission results and payload specifics) was taken from IBM course datasets (CSV files) and Wikipedia tables.

For uniformity, all of the data was combined into a single DataFrame.

For use in machine learning, EDA, visualization, and SQL queries, the dataset was locally stored as a CSV file.



Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts

1) Data Collection Process:

- Used Python's requests library to fetch launch data from SpaceX REST API v4
- Successfully retrieved **187 total launches** with HTTP status code 200
- Extracted **179 Falcon 9 launches** specifically for analysis
- API endpoint: <https://api.spacexdata.com/v4/launches/past>

2) Data Retrieved:

- **Launch Information:** Flight number, mission name, date/time (UTC), launch outcome
- **Rocket Details:** Rocket type, booster version, core information
- **Launch Sites:** Launchpad locations and identifiers
- **Payload Data:** Mass (kg), orbit type, customer information
- **Landing Outcomes:** Core landing status, landing location/method

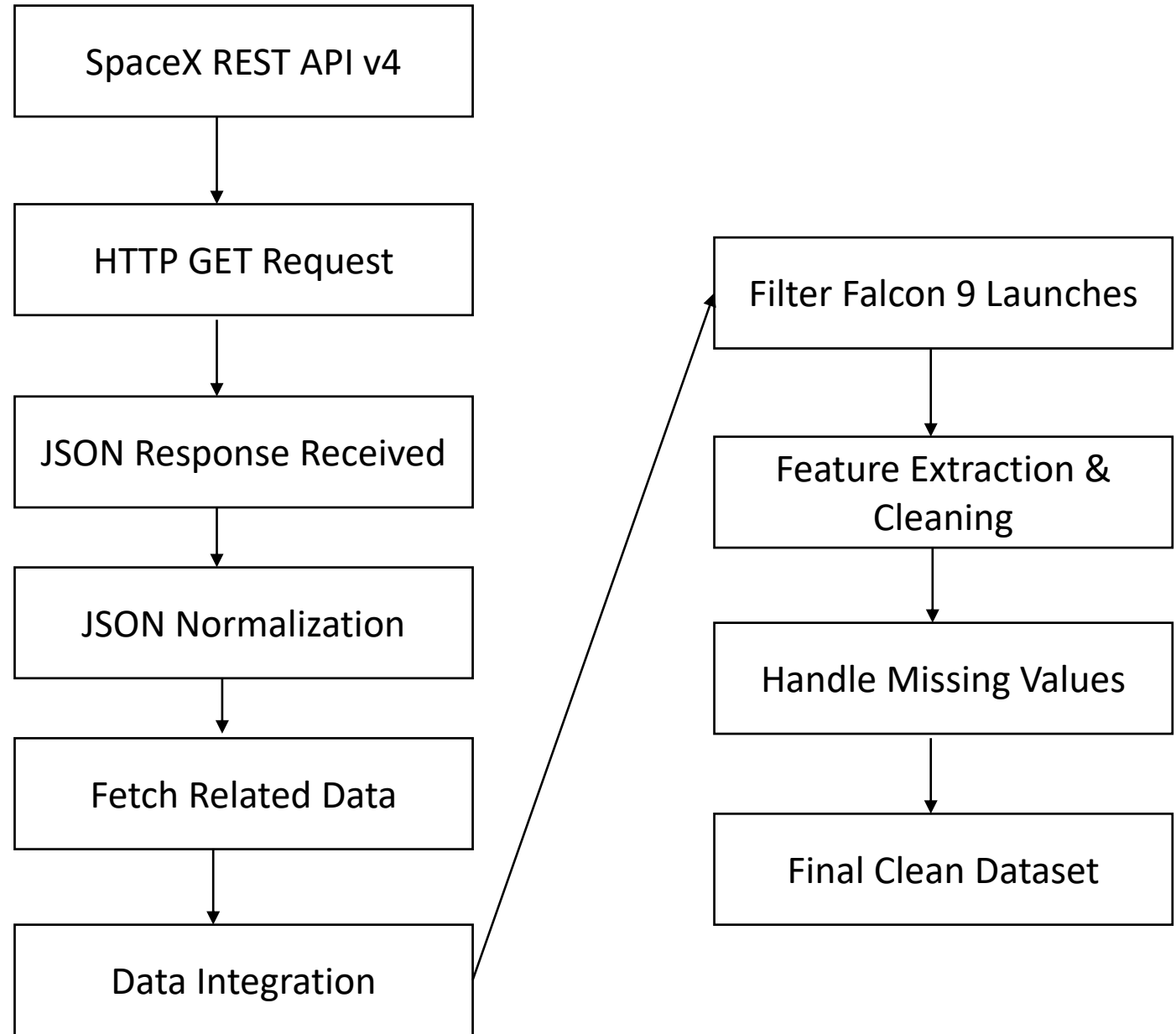
3) Data Processing Steps:

- JSON response normalized into tabular format using `pandas.json_normalize()`
- Multiple API endpoints queried:
 - Launches: 187 records
 - Rockets: 4 types
 - Launchpads: 6 locations
 - Payloads: 225 entries
 - Cores: 83 boosters
- **Date Range:** June 4, 2010 to October 5, 2022
- **Data Quality:**
 - Mission Success Rate: **98.88%**
 - Average Payload Mass: **8,117.57 kg**
 - Missing values handled: 23 in PayloadMass, 24 in LandingOutcome

GitHub URL of Data Collection with SpaceX REST calls:

https://github.com/sai-mounika-64/Data_Science_SpaceX_Capstone_Project/blob/main/spacex_data_collection_api.ipynb

Flowchart of Data Collection



Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts

Web Scraping process:

- **Data Extraction Phase**
 1. **HTTP GET Request:** Retrieve webpage content from Wikipedia
 2. **Status Code 200:** Successful HTTP response
 3. **HTML Parsing:** Convert raw HTML into navigable structure
 4. **BeautifulSoup4:** Python library for web scraping
- **Data Identification Phase**
 1. **Table Detection:** Locate <table> HTML elements
 2. **Header Extraction:** Identify column names from <th> tags
 3. **Row Parsing:** Extract data from <tr> and <td> tags
 4. **Multi-table Aggregation:** Combine 5 separate tables

- **Data Transformation Phase**

1. **Date Parsing:** Convert string dates to datetime objects
2. **Feature Engineering:** Extract year and month components
3. **Payload Mass Cleaning:** Remove units, convert to numeric
4. **Regular Expressions:** Pattern matching for data extraction
5. **Data Type Conversion:** String → Float/Int/DateTime

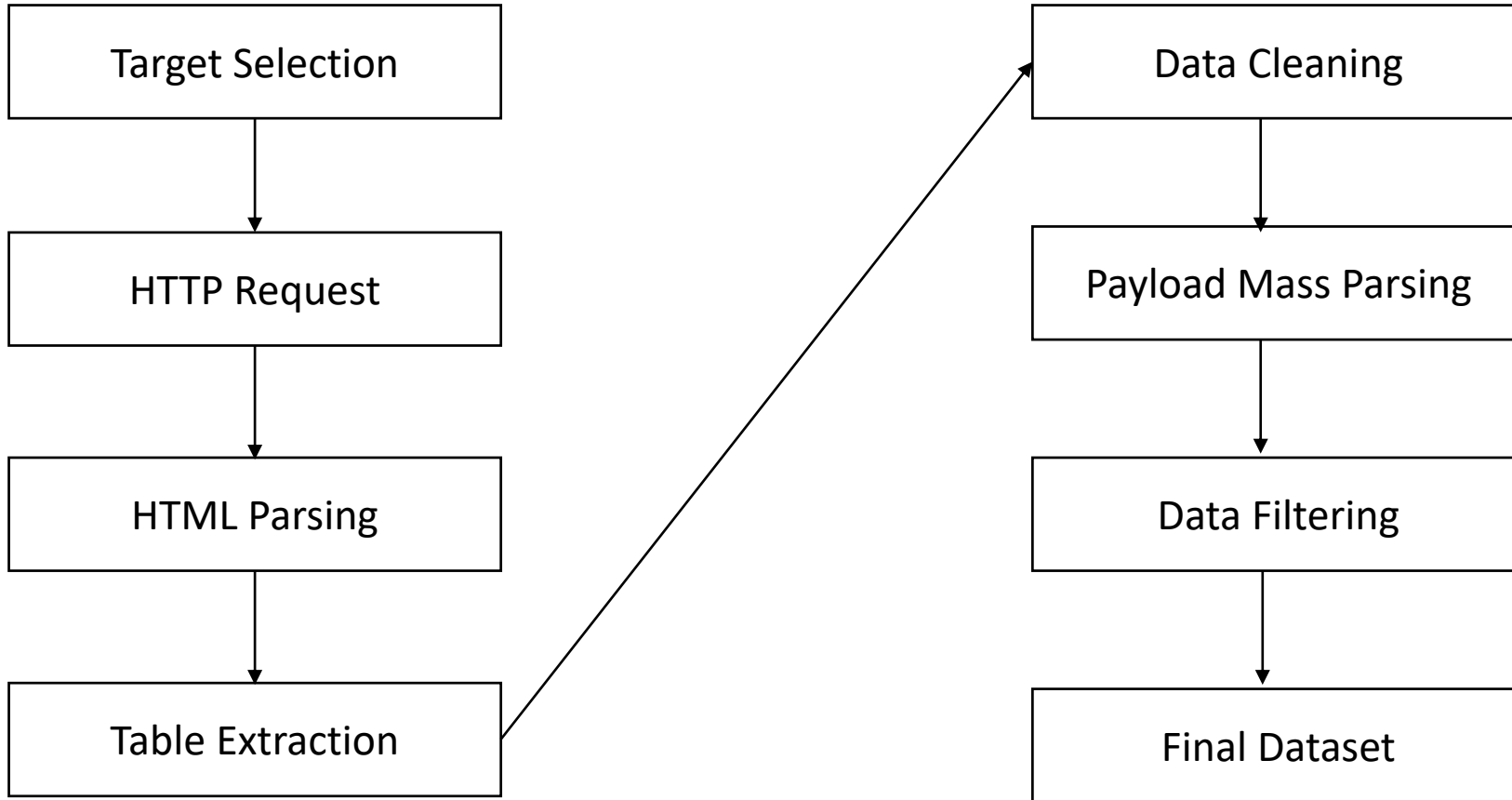
- **Data Quality Phase**

1. **Null Value Handling:** Remove incomplete records
2. **Data Validation:** Ensure required fields present
3. **Deduplication:** Remove description/annotation rows
4. **Data Reduction:** 699 → 39 high-quality records

GitHub URL of Web Scraping of Data

https://github.com/sai-mounika-64/Data_Science_SpaceX_Capstone_Project/blob/main/spacex_web_scraping.ipynb

Flowchart of Data Collection



Data Wrangling

- Describe how data were processed

1. Importing Required Libraries

- Imported essential libraries: **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn** for data manipulation and visualization.

2. Loading the Dataset

- Loaded the SpaceX launch dataset from a CSV file (spacex_launch_data.csv) into a Pandas DataFrame.
- Displayed dataset shape and column names to understand the data structure.

3. Initial Data Inspection

- Displayed the **first 10 rows** using `df.head(10)` to review data values and column types.
- Generated a **summary overview** of the dataset (data types, number of records, and feature list).

4. Handling Missing Values

- Checked for missing values using `isnull().sum()` and calculated the percentage of missing data per column.
- Visualized missing values using a horizontal **bar chart** for quick assessment.

5. Removing Duplicate Records

- Detected duplicate rows using `duplicated().sum()`.
- Removed duplicates using `drop_duplicates()` if found.

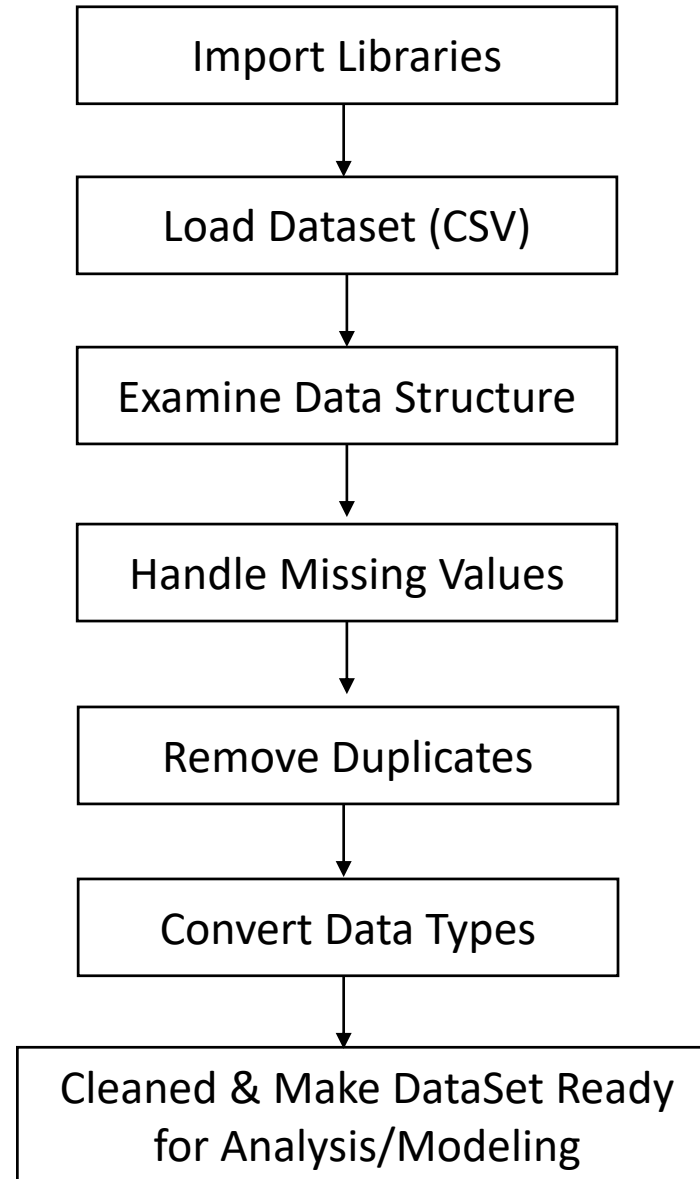
6. Data Type Conversion

- Converted Date column to **datetime** format.
- Ensured numeric columns (FlightNumber, PayloadMass, Year, Month) were of the correct numeric data types.

GitHub URL of Data Wrangling

https://github.com/sai-mounika-64/Data_Science_SpaceX_Capstone_Project/blob/main/spacex_data_wrangling.ipynb

Flowchart of Data Wrangling



EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts
 1. **Flight Number vs Launch Site (Scatter Plot)** - To show how the number of launches has increased across different launch sites. A scatter plot is ideal for visualizing relationships between numeric and categorical data
 2. **Payload Mass vs Launch Site (Scatter Plot)** - To analyze the relationship between payload mass and launch success across different sites. Helps identify if heavier payloads are more prone to failures or successes.
 3. **Payload Mass vs Launch Outcome (Scatter Plot with Regression)** - To explore whether payload mass affects landing success probability. Regression overlay helps quantify direction and strength of relationships, useful for predictive insights.
 4. **Flight Number vs Launch Outcome (Scatter Plot)** - To evaluate whether experience (flight number) contributes to higher success rates. Scatter plot reveals whether success rates improved as the number of launches increased.
 5. **Success Rate by Launch Site (Bar Chart)** - To compare success percentages among different launch sites. Bar charts are perfect for comparing categorical groups (launch sites) and visually showing which site performs best.
 6. **Launch Success Rate by Orbit Type (Bar Chart or Pie Chart)** - To show how success rates vary by orbit type (LEO, GTO, etc.). Summarizes categorical proportions and helps understand orbital mission complexity impact on outcomes.

7. Yearly Launch Trends (Line Chart or Bar Chart) - To visualize the number of launches per year and their success rates. Highlights temporal trends in launch frequency and performance over time.

8. Correlation Heatmap - To show relationships between multiple numerical features like PayloadMass, FlightNumber, and SuccessRate. Heatmaps provide a compact way to visualize correlation strength across several metrics simultaneously.

These charts collectively aimed to:

- Explore launch success patterns over time and across sites.
- Understand relationships among payload, orbit, and outcomes.

GitHub URL of EDA with Data Visualization

https://github.com/sai-mounika-64/Data_Science_SpaceX_Capstone_Project/blob/main/spacex_eda_visualization.ipynb

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
- **Created SQLite database**
 - Established connection to **spacex.db**
 - Loaded cleaned SpaceX dataset (**spacex_cleaned.csv**) into a SQL table named **SPACEXTBL**
- **Viewed database structure**
 - Used **PRAGMA table_info(SPACEXTBL);** to inspect table schema and confirm column names and data types.
 - Reviewed first 10 records to confirm correct data import and format.
 - Compared how frequently each site was used for launches.
 - Extracted only successful launch records for analysis.

- Retrieved failed launch data for comparison.
- Focused on medium-weight payloads to analyze their success rates.
- Checked success counts for each launch site.
- Determined the most frequently used orbits for missions.
- Examined growth of launches over time.
- Computed percentage of successful launches at each site.
- Found range of payload weights used in missions.

- **GitHub URL of EDA with Data Visualization**

https://github.com/sai-mounika-64/Data_Science_SpaceX_Capstone_Project/blob/main/spacex_eda_sql.ipynb

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
 - For Global Launch sites map, objects like **folium.Marker**, **folium.Popup**, **folium.Icon(color='blue', icon='rocket', prefix='fa')**, **folium.Circle** are added to visualize the global distribution of SpaceX launch sites and provide quick access to each site's information through interactive popups.
 - For Launch Outcomes Visualization map, objects like **folium.CircleMarker**, **folium.Popup**, **folium.Element(legend_html)** are used to analyze launch success rates and compare outcomes geographically across different sites.
 - For Proximity Analysis map, **folium.PolyLine** object is used along with **folium.Marker** and **folium.Circle** to draw lines connecting launch sites to nearby cities or recovery zones, showing spatial relationships and giving insight into logistical and operational factors.
 - For Timeline Map, **MarkerCluster** along with **folium.Marker** is used to group nearby markers together to make the map more readable. It focuses **on temporal exploration** of launch events

- **GitHub URL of Interactive Map with Folium**

https://github.com/sai-mounika-64/Data_Science_SpaceX_Capstone_Project/blob/main/spacex_interactive_folium.ipynb

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard.
Explain why you added those plots and interactions

- **Plots:**

1. **Success Pie Chart:**

- Built using `plotly.express.pie()`.
- Displays the proportion of successful vs failed launches across different launch sites.

2. **Payload vs. Launch Success Scatter Plot :**

- Built using `plotly.express.scatter()`.
- Plots payload mass on the x-axis and launch outcome (success/failure) on the y-axis.

- **Interactions:**

- **Dropdown :**

Allows users to filter results by launch site — updating both charts dynamically.

- **Range Slider :**

Lets users adjust the payload range, filtering scatter plot data interactively.

- **Callbacks:**

Used to link user input to graph updates, ensuring real-time interactivity.

These plots and interactions are added to

- Visually analyze **launch performance trends** across SpaceX sites.
- Explore the **relationship between payload mass and mission success**.
- Interactively **filter and compare** outcomes without manual data manipulation.

GitHub URL of ipynb file of Dashboard with Plotly Dash:

https://github.com/sai-mounika-64/Data_Science_SpaceX_Capstone_Project/blob/main/spacex_dash_app.ipynb

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model

1. Data Preparation:

- Data is split into training and testing sets using `train_test_split()`
- Features standardized and categorical variables encoded for model readiness.

2. Model Training:

The following models were developed:

- **Logistic Regression**
 - **Support Vector Machine (SVM)**
 - **Decision Tree Classifier**
 - **K-Nearest Neighbors (KNN)**
- Each model was tuned using **GridSearchCV** to test multiple hyperparameter combinations and selected the best model based on accuracy.

3. Model Evaluation:

- **classification_report()** was used to calculate accuracy, precision, recall, and F1-score.
- **Confusion matrices** were plotted and saved for visual comparison of model performance.

4. Model Improvement:

- **GridSearchCV** optimization improved model parameters (e.g., kernel, depth, neighbors).
- The best performing model was identified based on the highest accuracy and F1-score across all models.

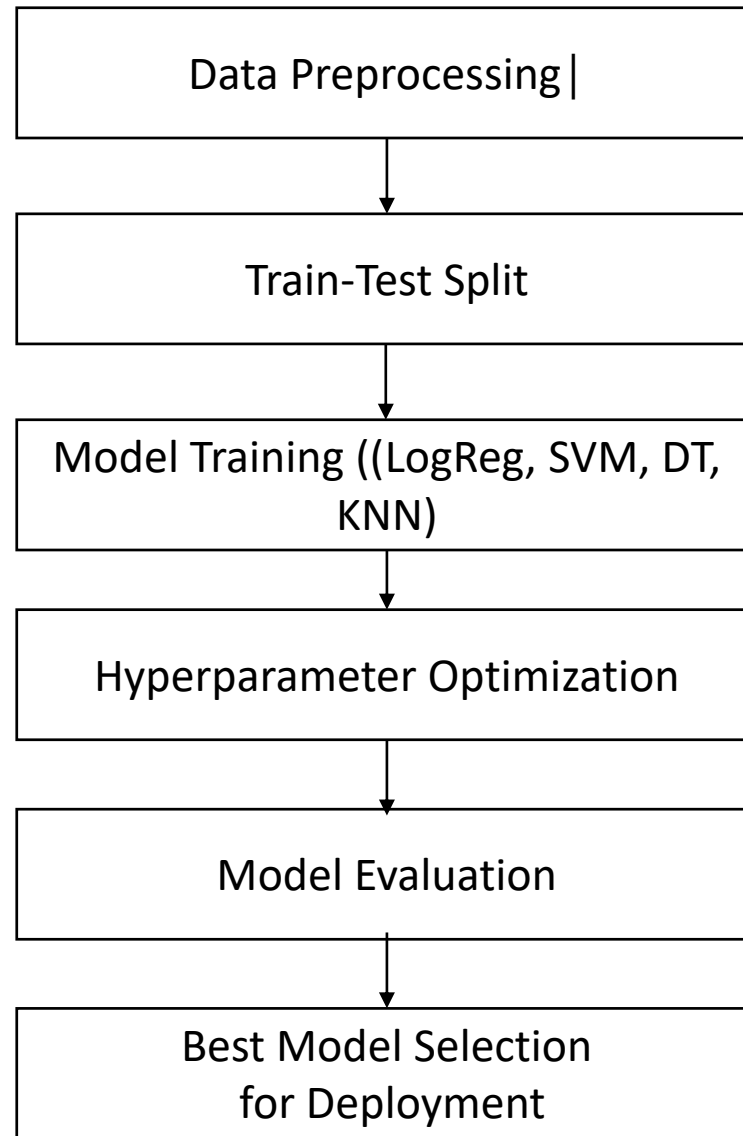
5. Final Model Selection:

- The best model's performance metrics and confusion matrix were stored as the final output.
- Visualization (plot_confusion_matrix) confirmed strong classification capability for launch success prediction.

GitHub URL of ipynb file of Predictive Analysis :

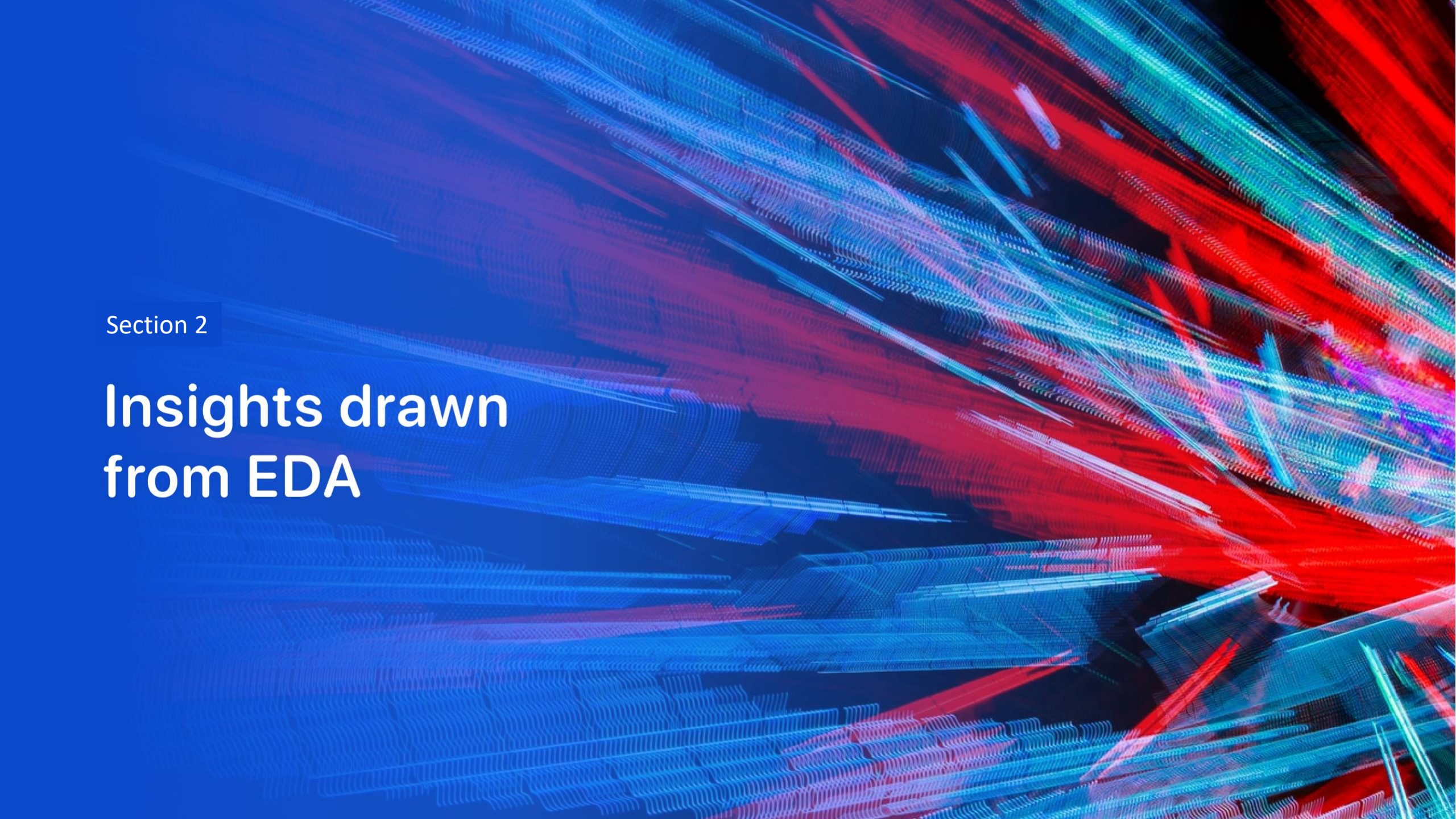
https://github.com/sai-mounika-64/Data_Science_SpaceX_Capstone_Project/blob/main/spacex_machine_learning.ipynb

Flowchart of Predictive Analysis



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

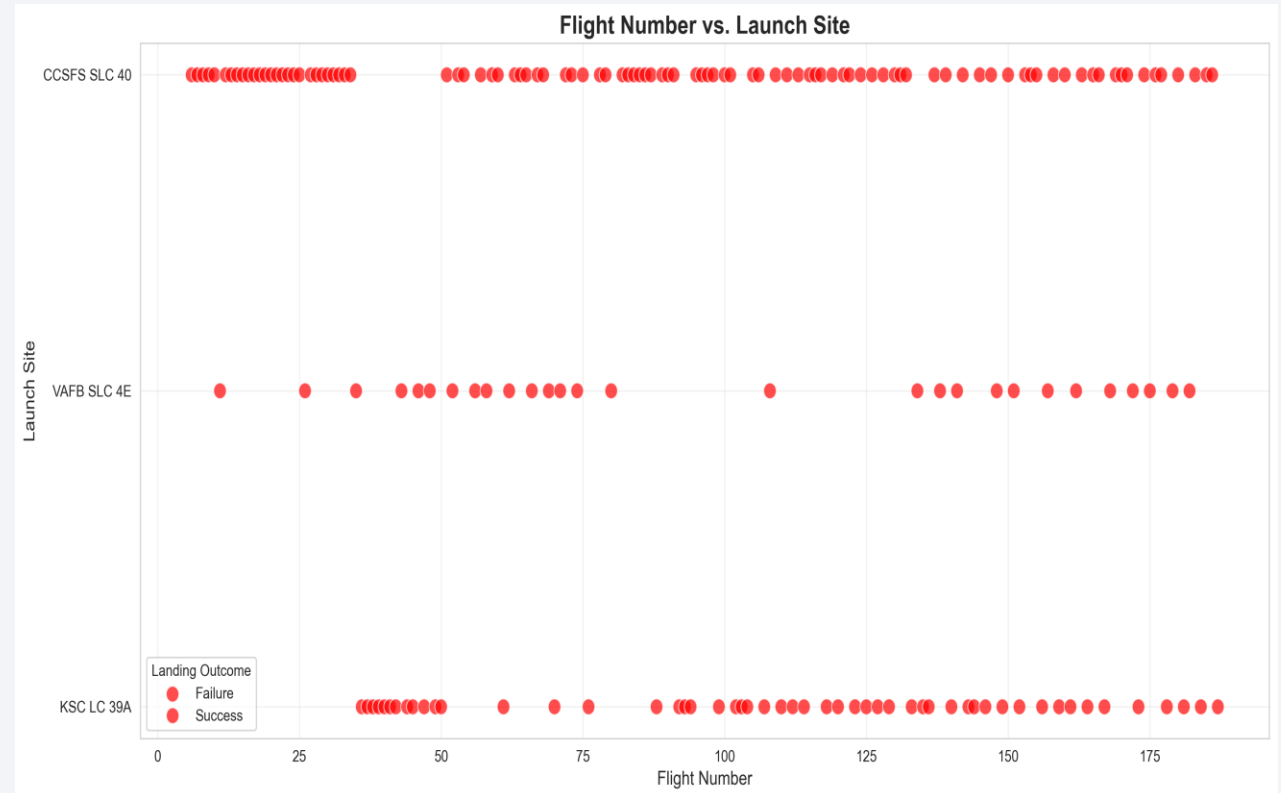
The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

Insights drawn from EDA

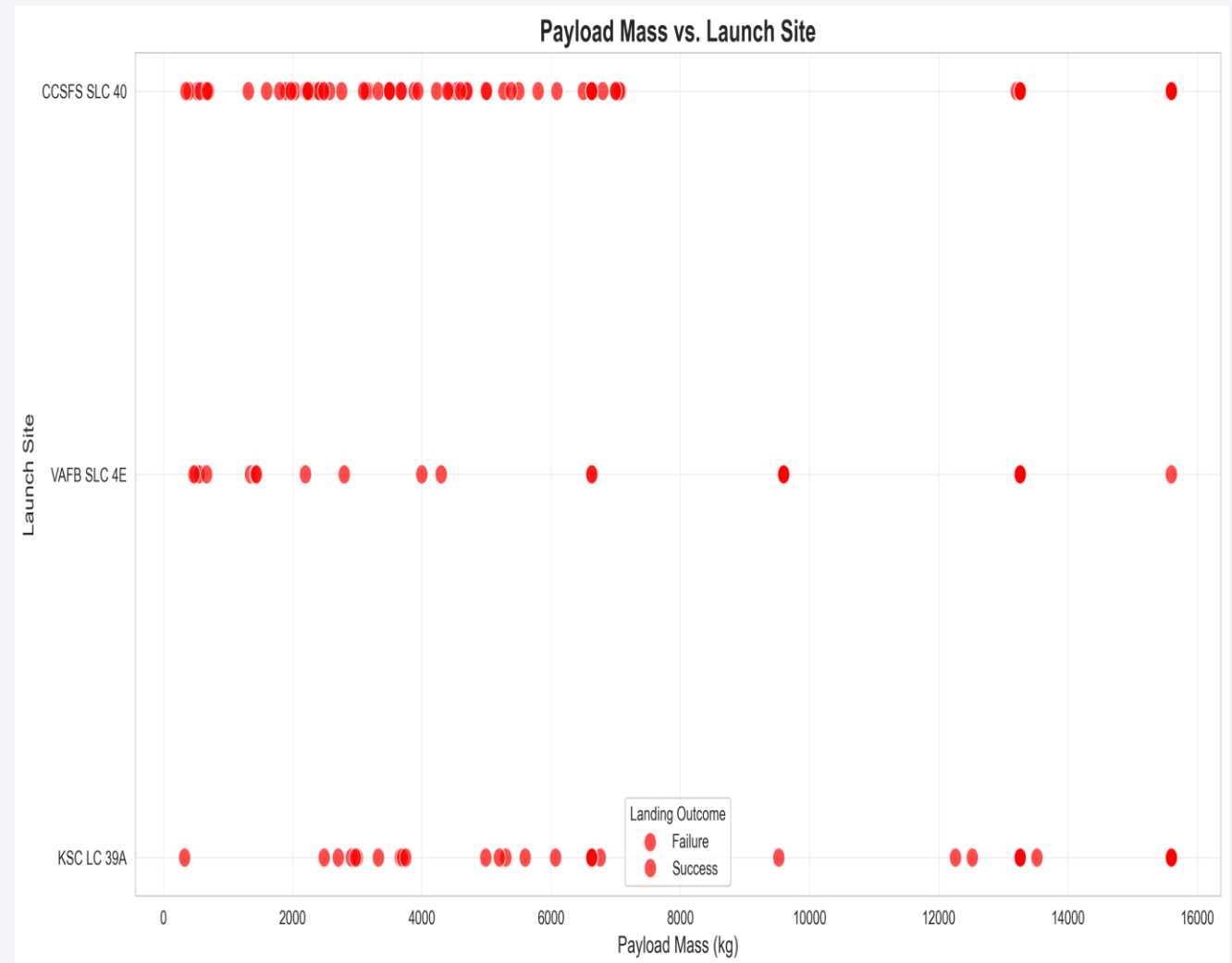
Flight Number vs. Launch Site

- Displays how launch success and failure vary across different SpaceX launch sites over increasing flight numbers.
- Helps identify which sites have higher success consistency as SpaceX gained experience.
- Indicates that experience and repeated launches led to higher success rates over time.



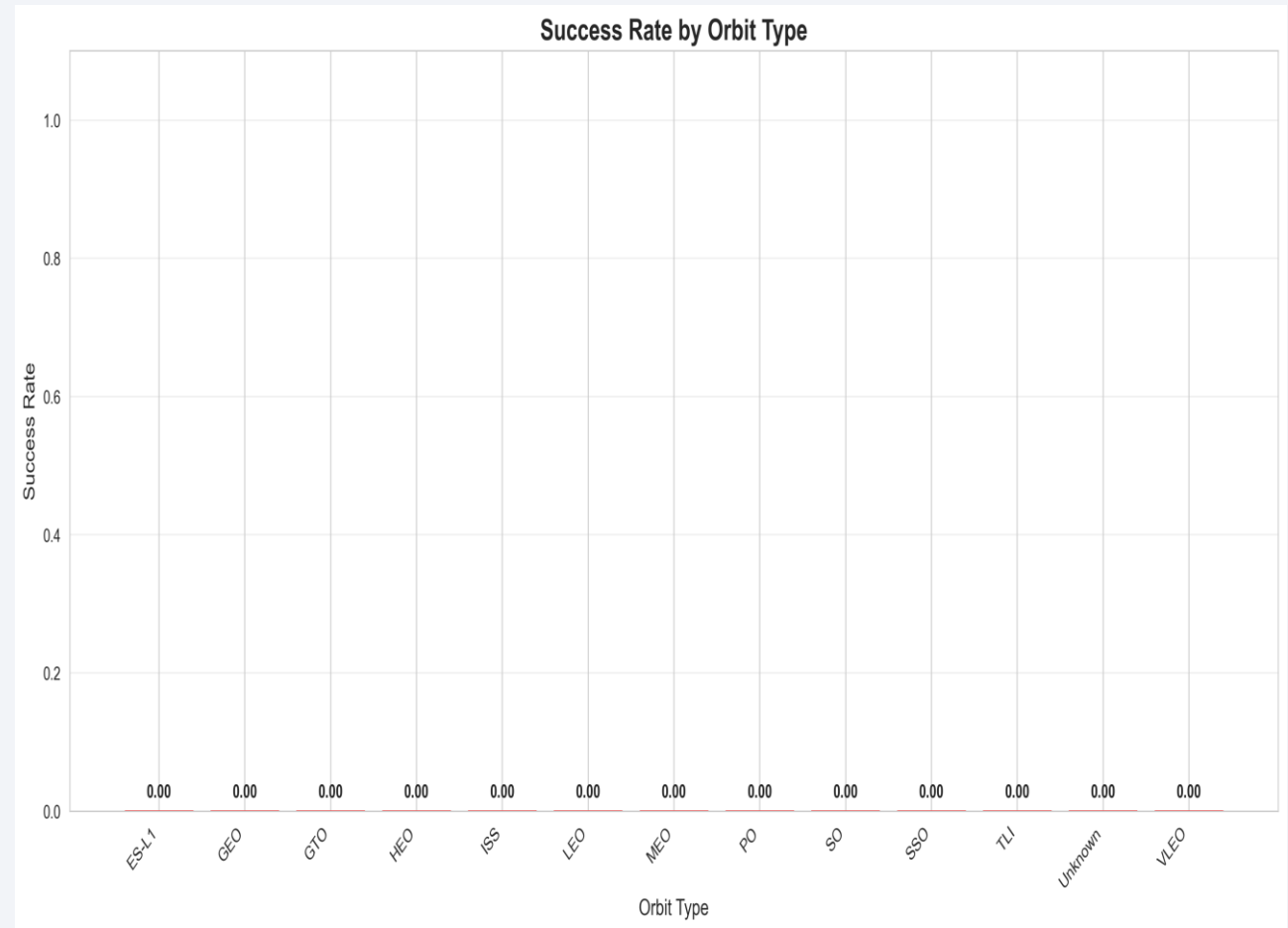
Payload vs. Launch Site

- Compares payload weights launched from each site to assess performance capacity.
- Shows how heavier payloads were distributed among launch sites.
- Helps determine if specific sites handled larger missions more reliably.



Success Rate vs. Orbit Type

- Summarizes the percentage of successful launches for each orbit category.
- Identifies which orbits have higher mission reliability.
- Provides insight into performance consistency across orbital destinations.



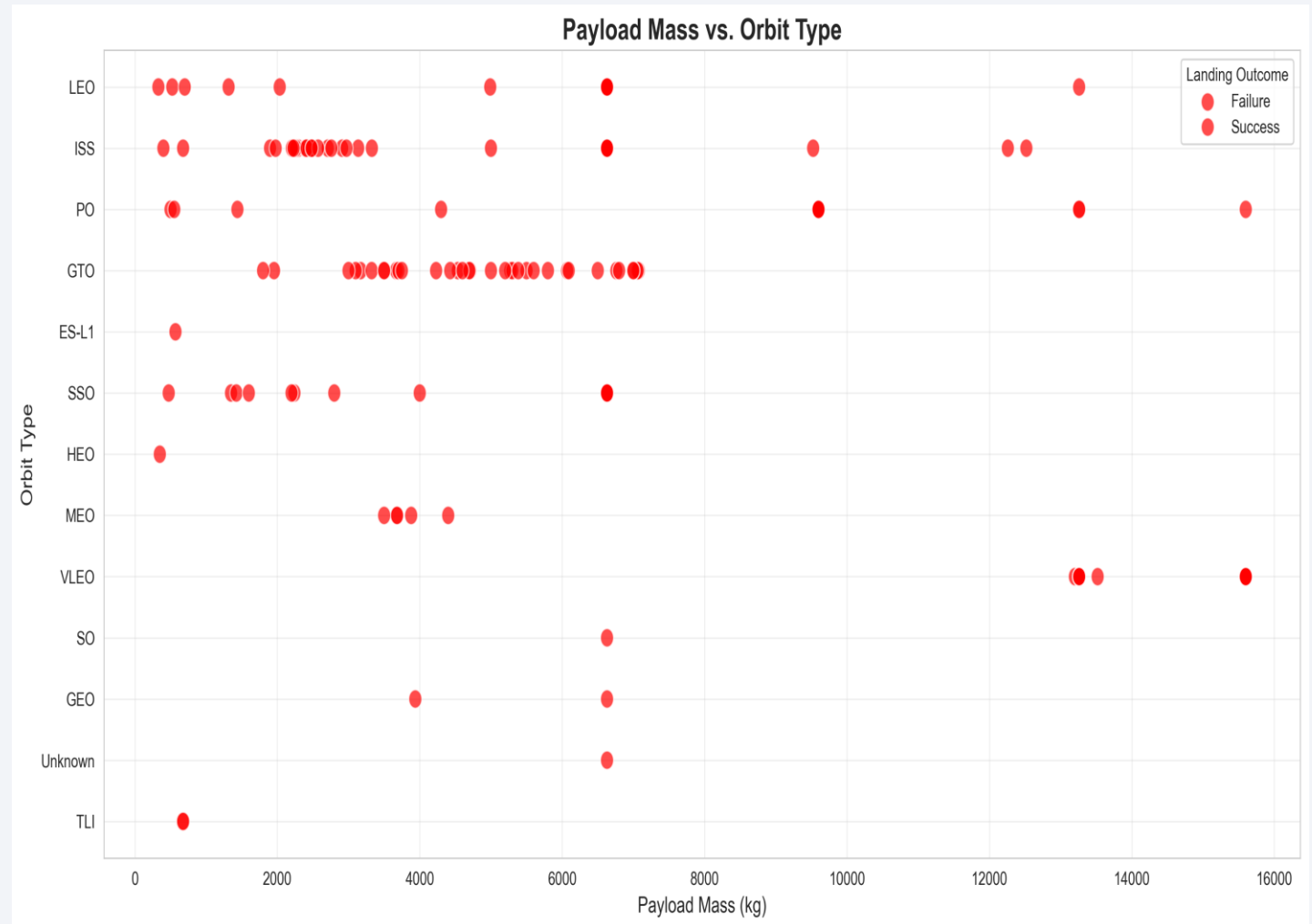
Flight Number vs. Orbit Type

- Displays the relationship between flight numbers and orbit destinations.
- Reveals how SpaceX expanded to more complex orbit types with time.
- Highlights performance patterns and success consistency across different orbits.



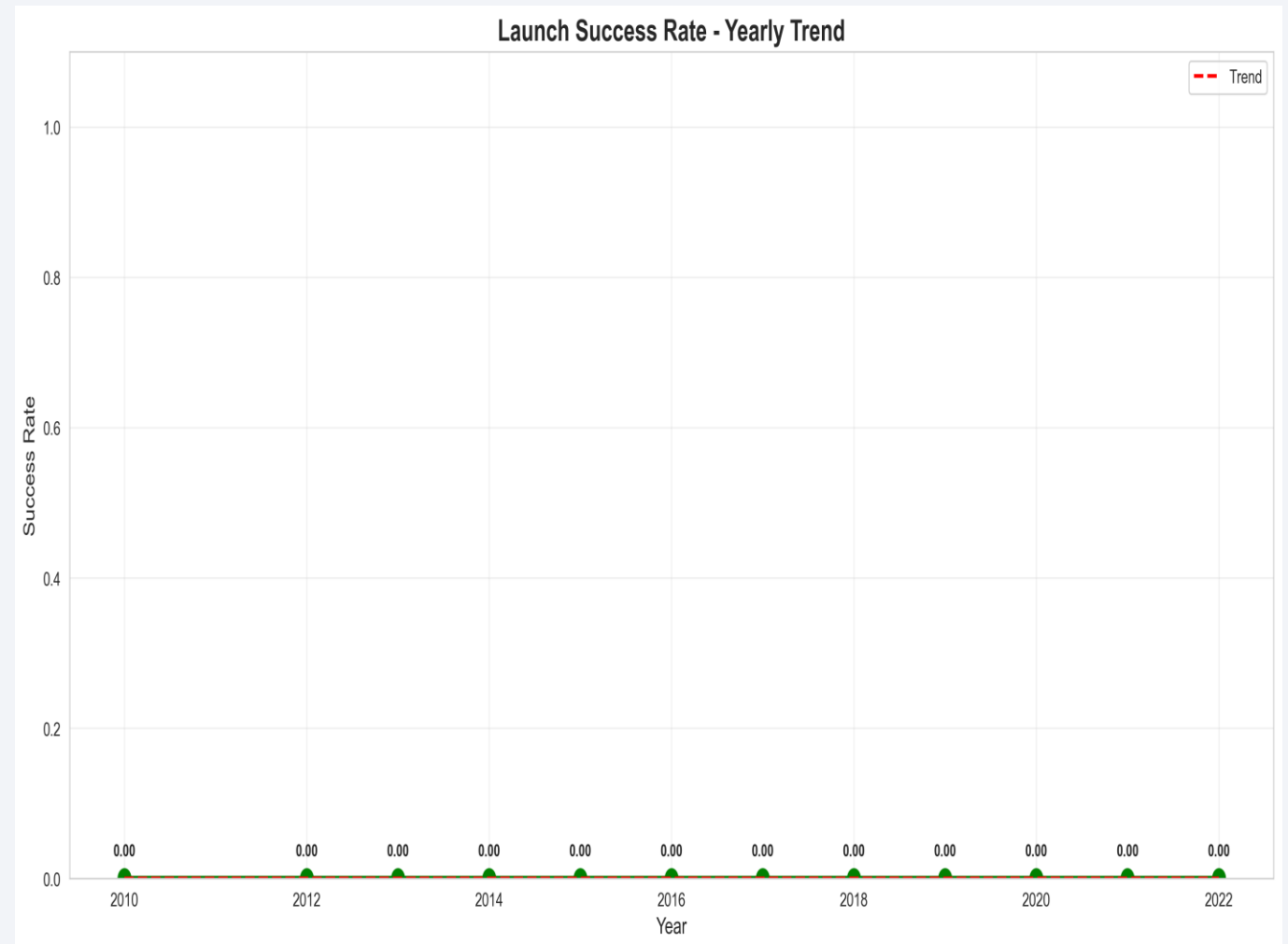
Payload vs. Orbit Type

- Illustrates how payload mass varies by orbit type and launch outcome.
- Highlights that certain orbits typically involve heavier payloads.
- Aids in analyzing success trends related to payload and orbit complexity.



Launch Success Yearly Trend

- Tracks annual SpaceX launch success rates over time.
- Shows clear improvement in success as technology and experience grew.
- Demonstrates SpaceX's steady progress toward mission reliability and efficiency.



All Launch Site Names

- Find the names of the unique launch sites

Query

```
SELECT DISTINCT LaunchSite FROM SPACEXTBL ORDER BY LaunchSite;
```

Result

Unique Launch Sites:	
	LaunchSite
0	CCSFS SLC 40
1	KSC LC 39A
2	VAFB SLC 4E
Total unique launch sites: 3	

There are **three unique SpaceX launch sites**: two in Florida (CCSFS and KSC) and one in California (VAFB).

Launch Site Names Begin with 'CCS'

- Find 5 records where launch sites begin with 'CCS'

Query

```
SELECT * FROM SPACEXTBL WHERE  
LaunchSite LIKE 'CCS%' LIMIT 5;
```

Result

- All five records belong to **Cape Canaveral Air Force Station (LC-40)**, confirming the pattern search.

5 records where launch site begins with 'CCS':

	FlightNumber	Date	Year	Month	Quarter	LaunchSite	PayloadMass	Payload_Category	Orbit	Orbit_clean	MissionOutcome	LandingOutcome	Class
0	6	2010-06-04 18:45:00+00:00	2010	6	2	CCSFS SLC 40	6630.5	Heavy	LEO	LEO	1	No attempt	0
1	7	2010-12-08 15:43:00+00:00	2010	12	4	CCSFS SLC 40	6630.5	Heavy	LEO	LEO	1	No attempt	0
2	8	2012-05-22 07:44:00+00:00	2012	5	2	CCSFS SLC 40	525.0	Light	LEO	LEO	1	No attempt	0
3	9	2012-10-08 00:35:00+00:00	2012	10	4	CCSFS SLC 40	400.0	Light	ISS	ISS	1	No attempt	0
4	10	2013-03-01 19:10:00+00:00	2013	3	1	CCSFS SLC 40	677.0	Light	ISS	ISS	1	No attempt	0

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

Query

```
SELECT SUM(PayloadMass) as Total_Payload_Mass FROM SPACEXTBL WHERE  
MissionOutcome LIKE '%CRS';
```

Results:

Displays Total Payload Mass as 15600

- Boosters carrying NASA payloads have transported a **total of 15,600 kg** of payload mass.

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

Query

```
SELECT    AVG(PayloadMass) as Average_Payload_Mass, COUNT(*) as  
Mission_Count, MIN(PayloadMass) as Min_Payload, MAX(PayloadMass) as  
Max_Payload FROM SPACEXTBL;
```

Results:

Overall payload mass statistics:

	Average_Payload_Mass	Mission_Count	Min_Payload	Max_Payload
0	7926.497486	179	330.0	15600.0

- The **F9 v1.1** boosters carried an average payload mass of 7926.4 kg, indicating medium-capacity missions.

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

Query

```
SELECT MIN(Date) as First_Success_Date, FlightNumber, LaunchSite, LandingOutcome FROM SPACEXTBL  
WHERE LandingOutcome LIKE '%success%' AND (LandingOutcome LIKE '%ground%' OR LandingOutcome LIKE  
'%rtls%') GROUP BY FlightNumber, LaunchSite, LandingOutcome ORDER BY DateLIMIT 1;
```

- The first successful ground pad landing occurred on December 22, 2015, marking a milestone in SpaceX recovery efforts.

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Query

```
SELECT FlightNumber, PayloadMass, LandingOutcome, Date, LaunchSite FROM SPACEXTBL WHERE  
LandingOutcome LIKE '%success%' AND (LandingOutcome LIKE '%drone%' OR LandingOutcome LIKE '%asds%' OR  
LandingOutcome LIKE '%ocean%') AND PayloadMass > 4000 AND PayloadMass < 6000 ORDER BY Date;
```

Results:

The boosters are : F9 FT B1021 and F9 B1031

- Two booster versions (**F9 FT B1021** and **F9 B1031**) met the criteria of successful drone ship landings within the given payload range.

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

Query

```
SELECT Class, COUNT(*) as Count, ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM SPACEXTBL), 2) as Percentage FROM SPACEXTBL GROUP BY Class ORDER BY Class;
```

Results:

Mission outcomes summary:

	Class	Count	Percentage
0	0	35	19.55
1	1	144	80.45

Labeled outcomes:

	Outcome	Count	Percentage
0	Failure	35	19.55
1	Success	144	80.45

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

Query

```
SELECT MAX(PayloadMass) as Max_Payload FROM SPACEXTBL;
```

Results:

Top 10 heaviest payloads:						
	FlightNumber	PayloadMass	Date	LaunchSite	Class	
0	84	15600.0	2019-11-11 14:56:00+00:00	CCSFS SLC 40	0	
1	87	15600.0	2020-01-07 02:19:00+00:00	CCSFS SLC 40	0	
2	89	15600.0	2020-01-29 14:06:00+00:00	CCSFS SLC 40	0	
3	90	15600.0	2020-02-17 15:05:55+00:00	CCSFS SLC 40	0	
4	92	15600.0	2020-03-18 12:16:00+00:00	KSC LC 39A	0	
5	93	15600.0	2020-04-22 19:30:00+00:00	KSC LC 39A	0	
6	95	15600.0	2020-06-04 01:25:00+00:00	CCSFS SLC 40	0	
7	96	15600.0	2020-06-13 09:21:00+00:00	CCSFS SLC 40	0	
8	99	15600.0	2020-08-07 05:12:00+00:00	KSC LC 39A	0	
9	100	15600.0	2020-08-18 14:31:00+00:00	CCSFS SLC 40	0	

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Query

```
SELECT FlightNumber, PayloadMass, Date, LaunchSite, Orbit_clean as Orbit,  
Class FROM SPACEXTBL WHERE PayloadMass = (SELECT MAX(PayloadMass) FROM  
SPACEXTBL);
```

Results:

Failed drone ship landings in 2015:

	Date	FlightNumber	LaunchSite	LandingOutcome	PayloadMass
0	2015-01-10 09:47:00+00:00	19	CCSFS SLC 40	ASDS	2395.0
1	2015-02-11 23:03:00+00:00	20	CCSFS SLC 40	Ocean	570.0
2	2015-04-14 20:10:00+00:00	22	CCSFS SLC 40	ASDS	1898.0
3	2015-06-28 14:21:00+00:00	24	CCSFS SLC 40	ASDS	2477.0

In **2015**, there were 4 **failed drone ship landings**, performed by **F9 v1.1**

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Query

```
SELECT FlightNumber, PayloadMass, Date, LaunchSite, Class FROM  
SPACEXTBL ORDER BY PayloadMass DESC LIMIT 10;
```

Results:

```
=====
TASK 10: Ranked Landing Outcomes (2010-06-04 to 2017-03-20)
=====
```

Landing outcomes ranked by count:

	LandingOutcome	Count
0	No attempt	12
1	ASDS	12
2	Ocean	5
3	RTLS	3

With percentages:

	LandingOutcome	Count	Percentage
0	No attempt	12	37.50
1	ASDS	12	37.50
2	Ocean	5	15.62
3	RTLS	3	9.38

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Global Launch Sites View

Elements:

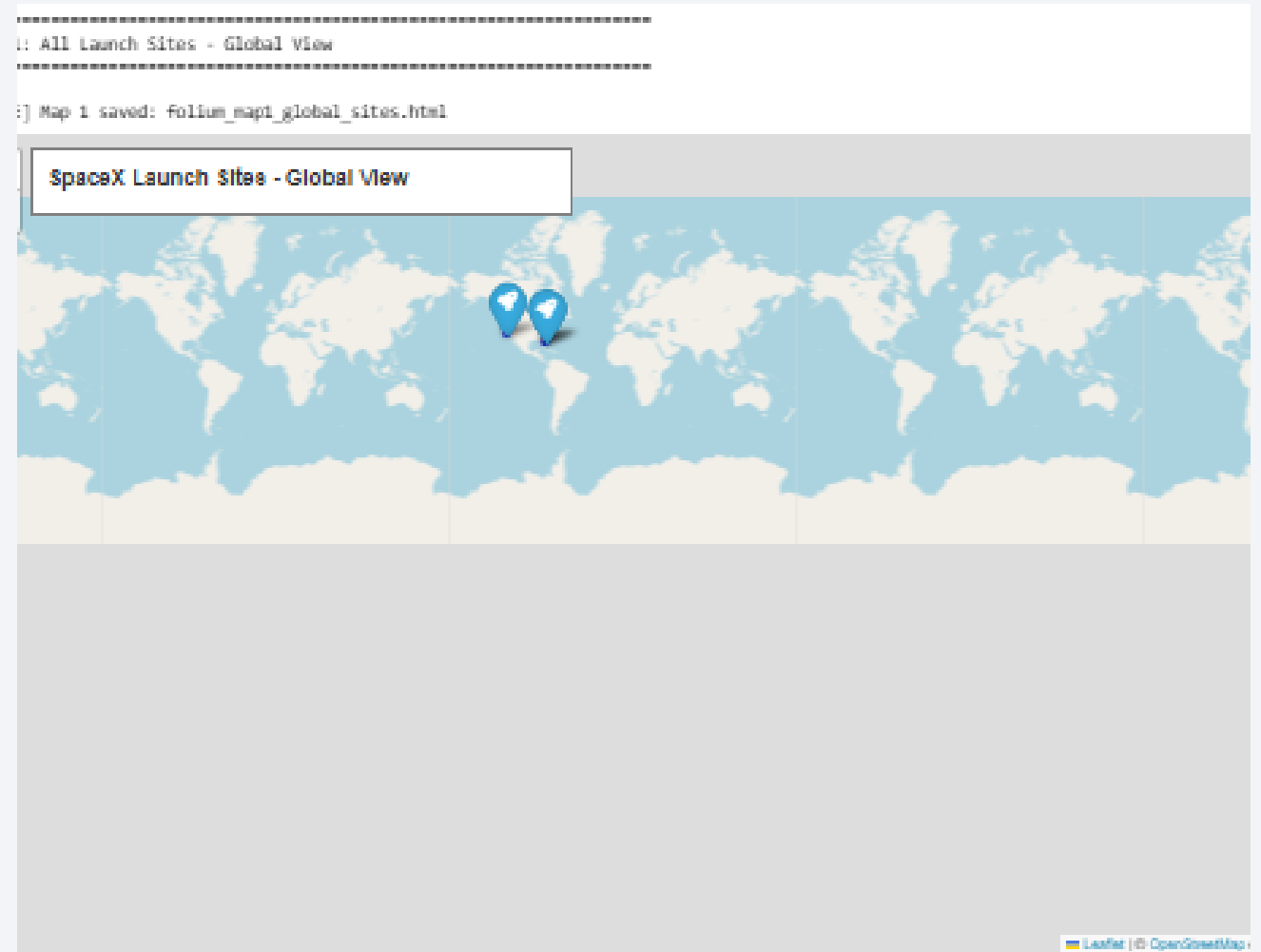
Blue markers (rocket icons): Represent SpaceX launch site locations.

Popups: Show site name, number of launches, successful launches, and success rate.

Blue circles: Indicate launch frequency (radius proportional to number of launches).

Title box: “SpaceX Launch Sites - Global View” fixed at the top for context.

- To visualize where SpaceX launch sites are geographically located and compare their activity levels.



Launch Outcomes

Elements:

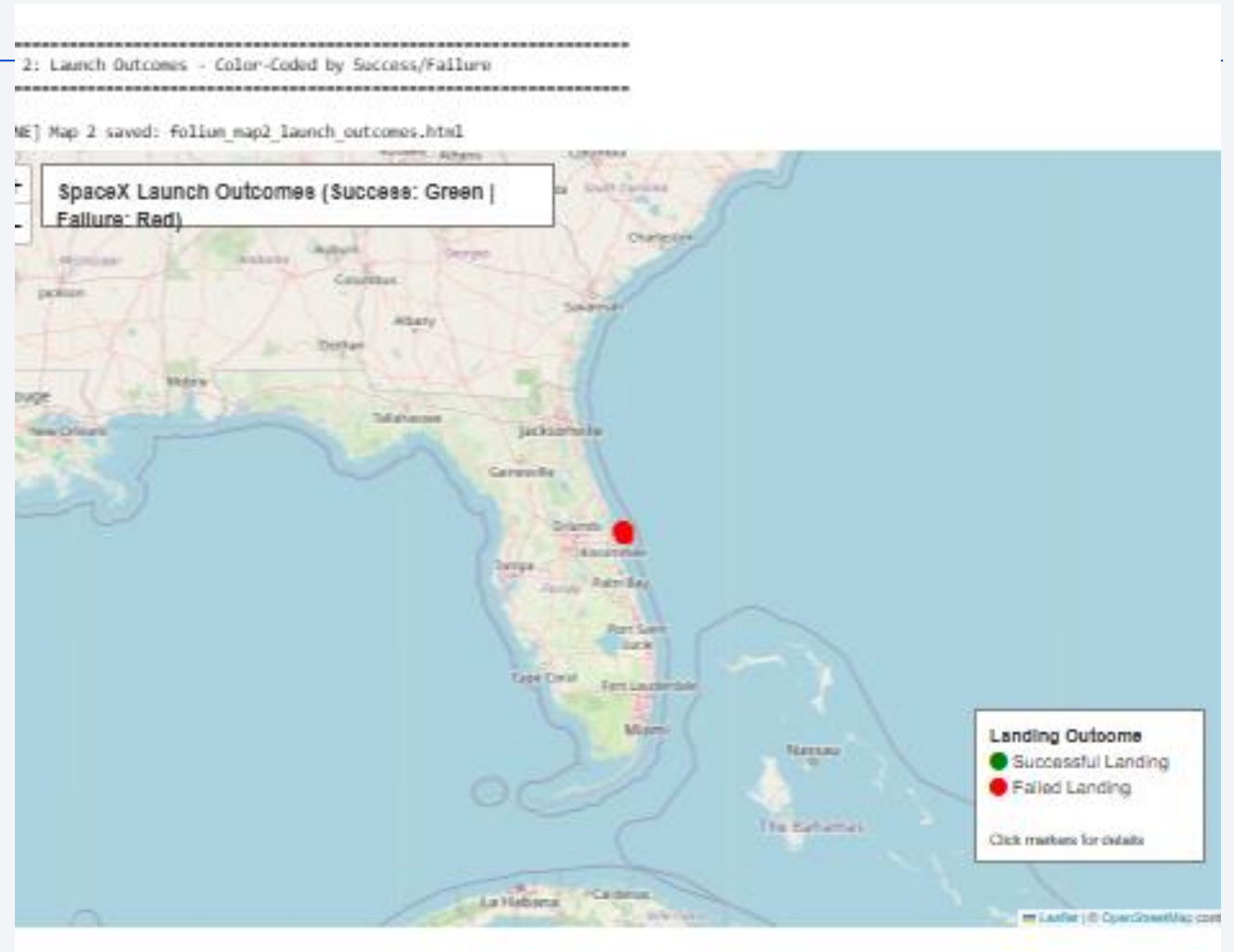
Green and red circular markers: Represent successful (green) and failed (red) launches.

Popups: Include flight number, date, site, payload, orbit, and outcome.

Legend: Explains color meaning (success vs. failure).

Title box: Describes the map theme for quick reference.

- To analyze success and failure outcomes of launches geographically and understand performance at each site.



KSC Proximities and Distances

Elements:

Red marker: Kennedy Space Center (KSC LC-39A) launch site.

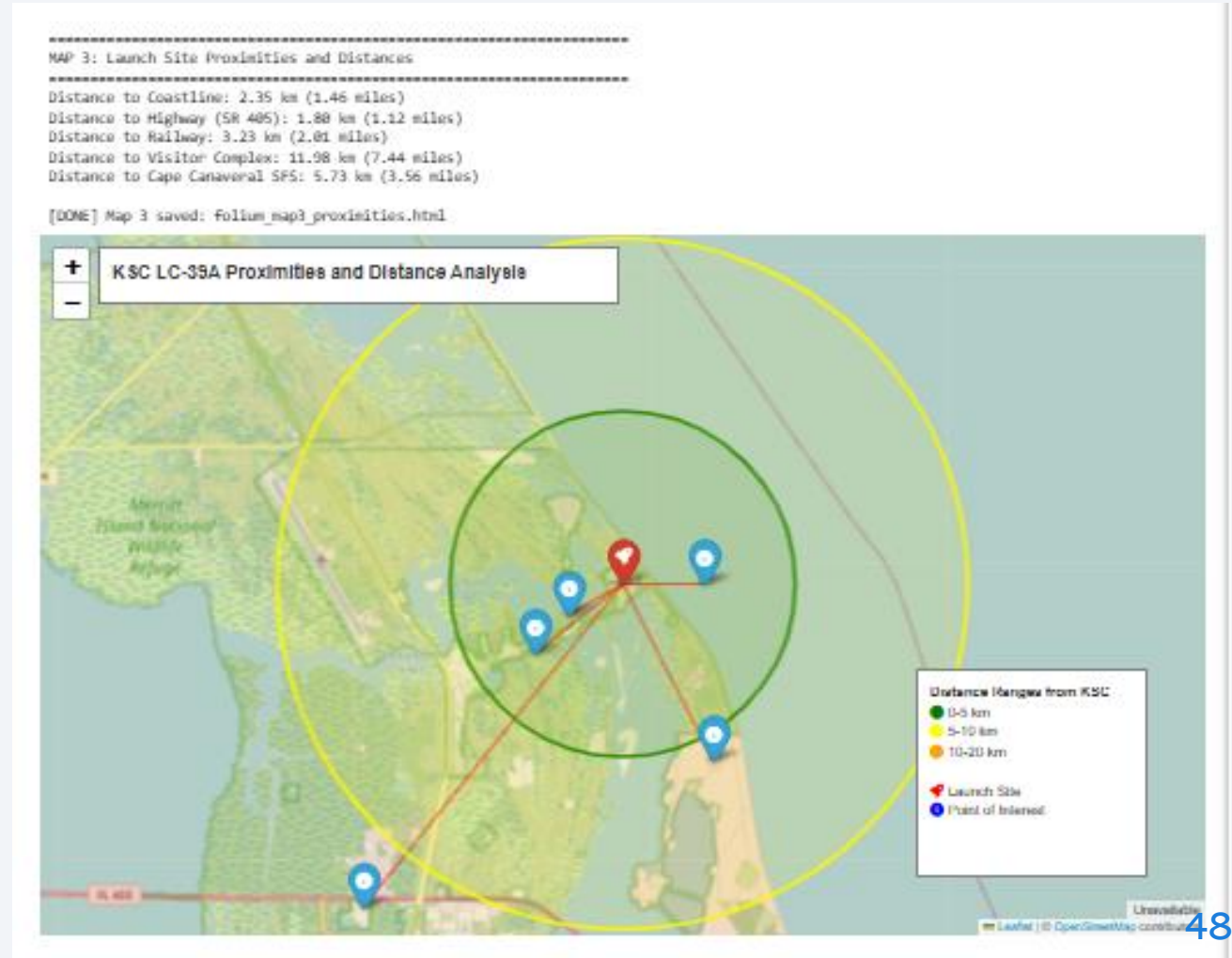
Blue markers: Nearby points of interest

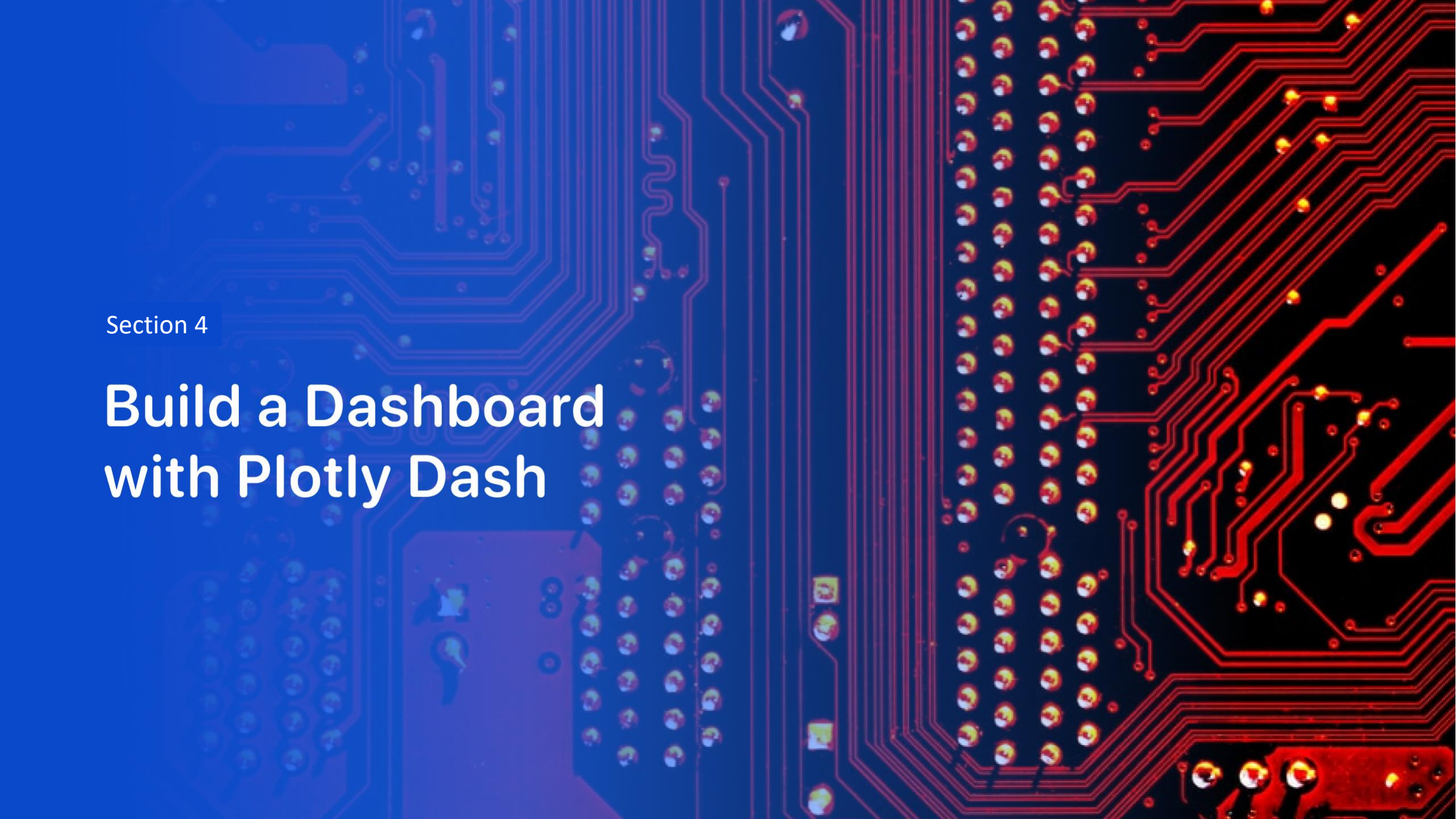
Red lines: Show distance connections between KSC and those points.

Concentric colored circles: Indicate 5 km (green), 10 km (yellow), and 20 km (orange) distance ranges.

Legend: Explains color coding for distance zones and icons.

- To analyze proximity relationships between the launch site and nearby infrastructure for safety and logistics insights.



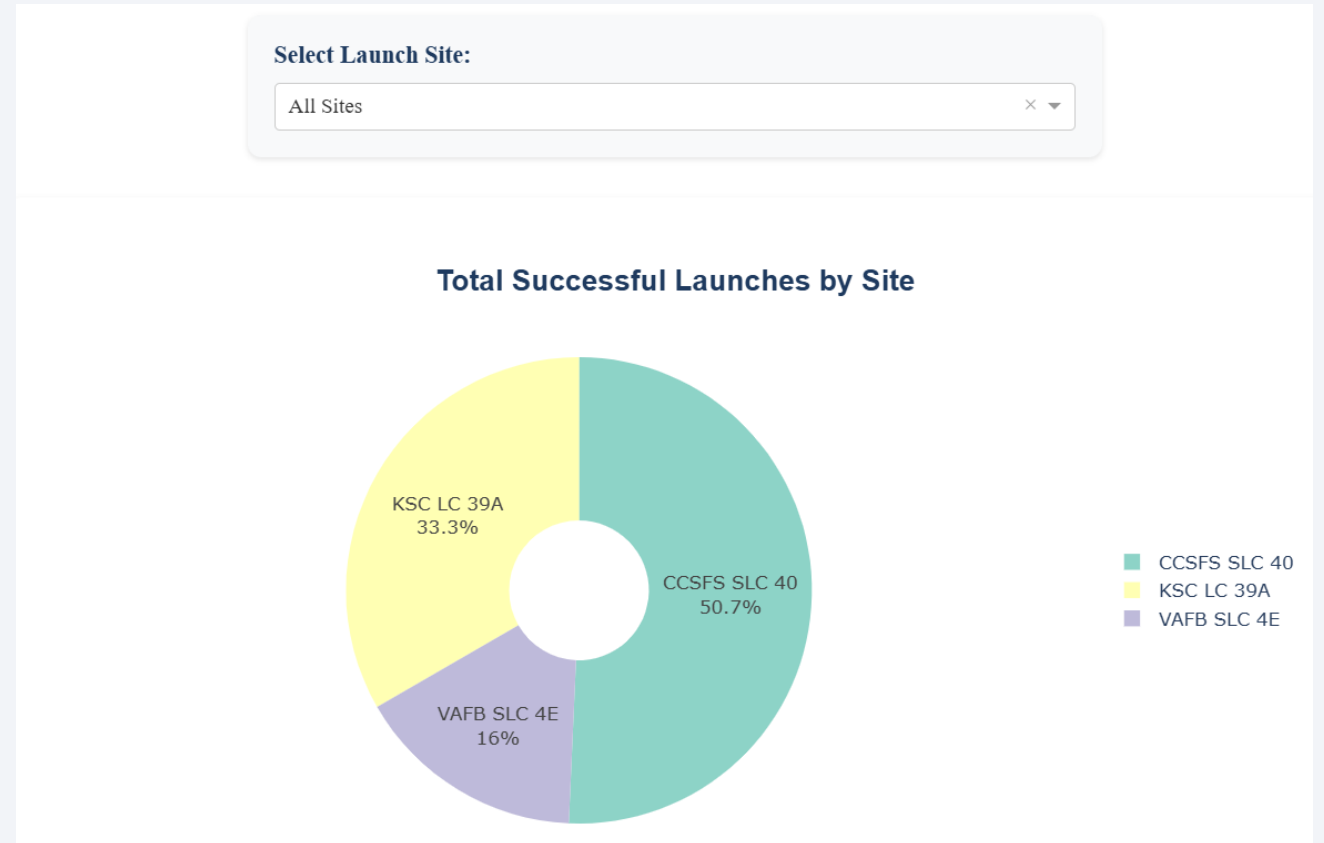


Section 4

Build a Dashboard with Plotly Dash

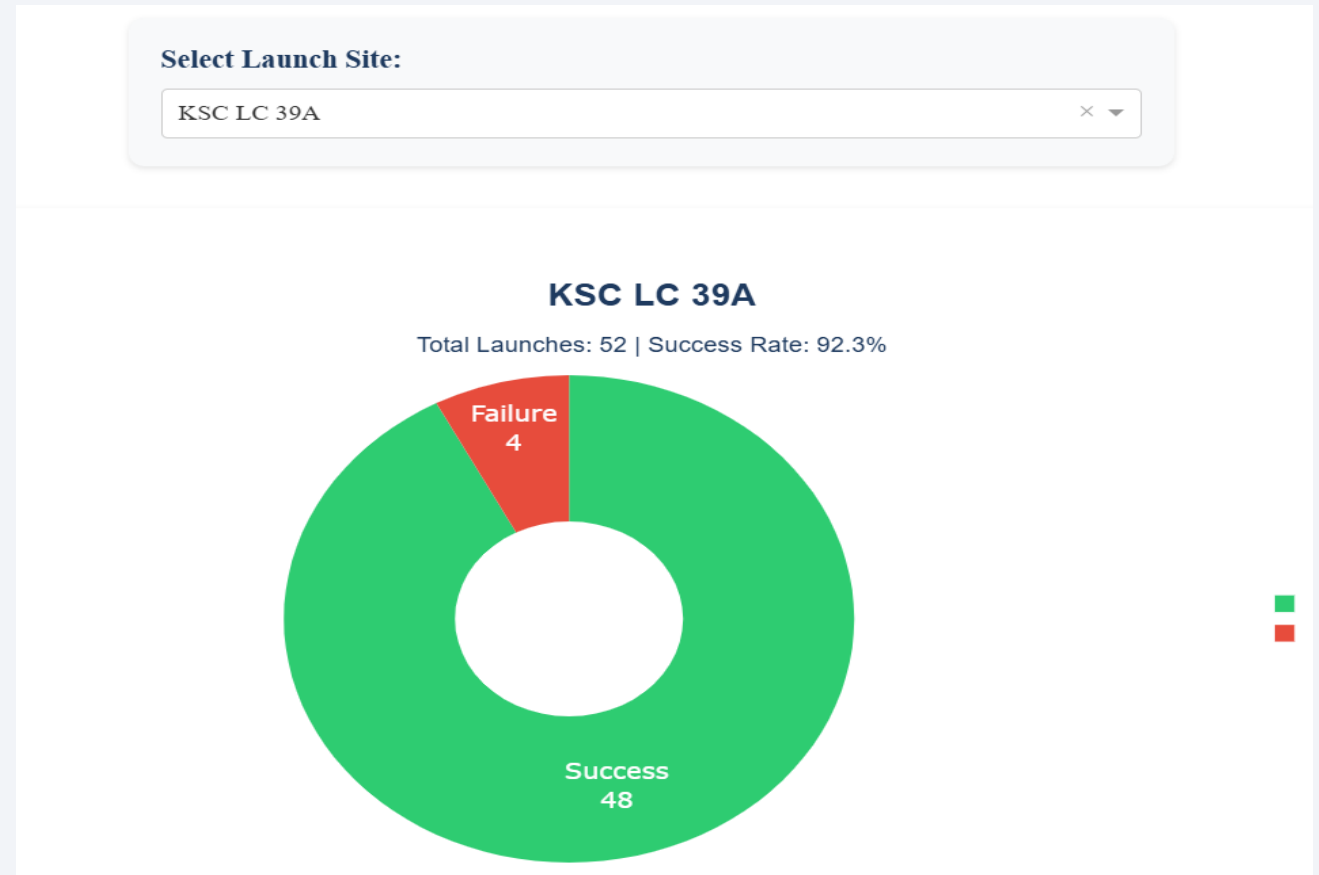
Total Successful Launches by all sites

- This chart shows the proportion of successful launches from all SpaceX launch sites combined.
- Each slice of the pie corresponds to one launch site.
- Share of each Launch Site for Total Successful Launches
 - CCSFS SLC 40 - 50.7%
 - KSC LC 39A - 33.3%
 - VAFB SLC 4E – 16%



Launch site with highest launch success ratio

- This chart is a drill-down for a single site (KSC LC 39A) which is having the highest launch success ratio.
- It compares successful and failed launches from that site.
- High success rate indicates KSC LC 39A is a reliable and mature launch site.
- The other two sites success ratios are:
 - CCSFS SLC 40 - Total Launches – 99, Success Rate - 73.7%
 - VAFB SLC 4E - Total Launches - 28, Success Rate - 82.1%



Payload vs. Launch Outcome scatter plot for all sites

- The success rate increases with payload mass not because heavy payloads are easier, but because they correspond to newer rocket versions (like Falcon 9 Block 5 and Falcon Heavy) which are far more reliable than earlier models.
- The payload pattern gives us clues:
 - Lower payloads (0–2 t) likely correspond to Falcon 1 or early Falcon 9 v1.0.
 - Mid-range (2–8 t) missions are typical of Falcon 9 Full Thrust variants.
 - High payloads (8–16 t) correspond to Falcon 9 Block 5 or Falcon Heavy, both having near-perfect success rates.

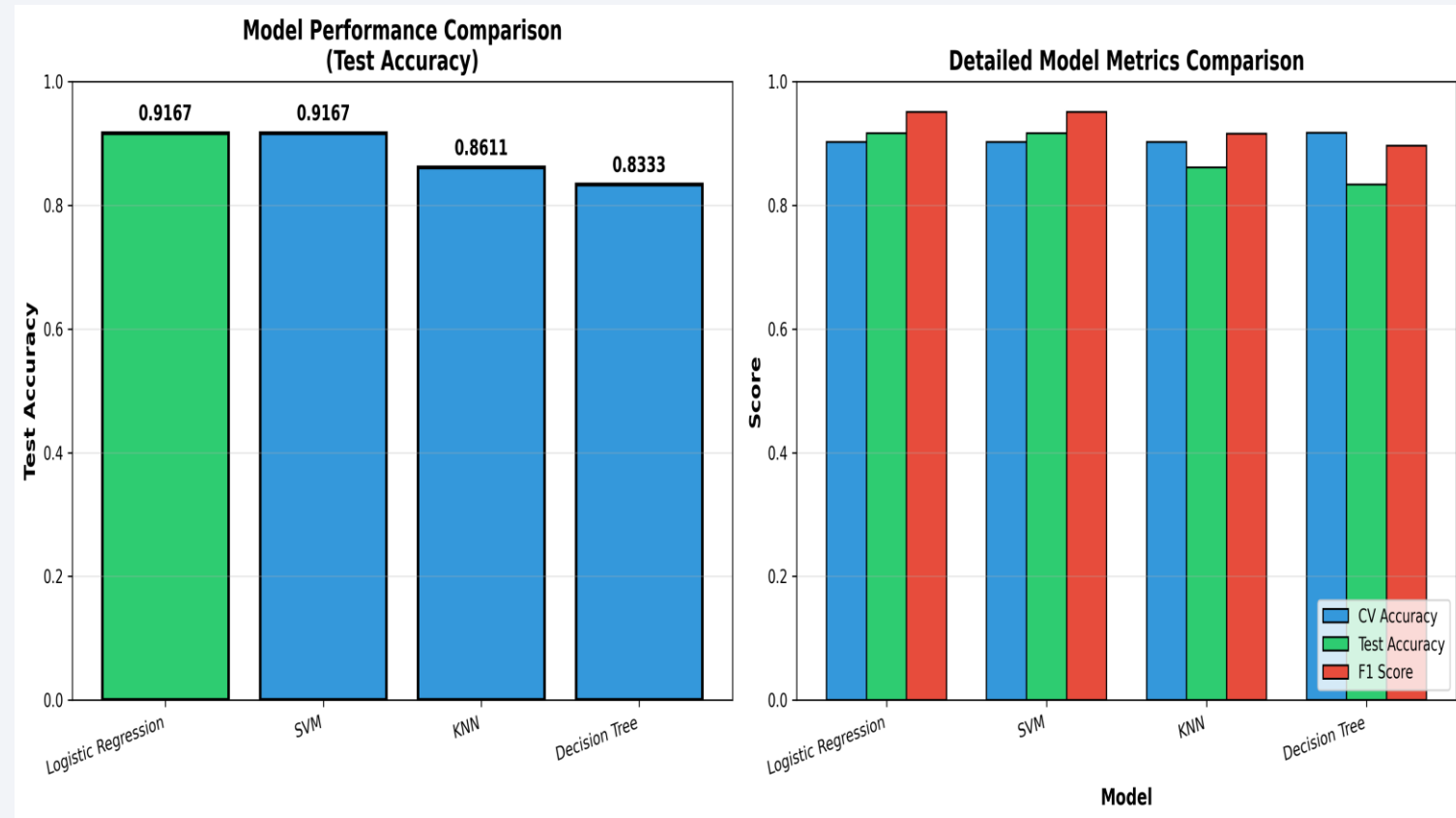


Section 5

Predictive Analysis (Classification)

Classification Accuracy

- From the bar chart (Model Performance Comparison), both Logistic Regression and SVM models achieved the highest test accuracy of 0.9167. But, Logistic Regression is considered the best overall model because of its consistent performance across all evaluation metrics.
- The model correctly predicted 29 successful landings, showing strong reliability in detecting successful outcomes and there are no false negatives.

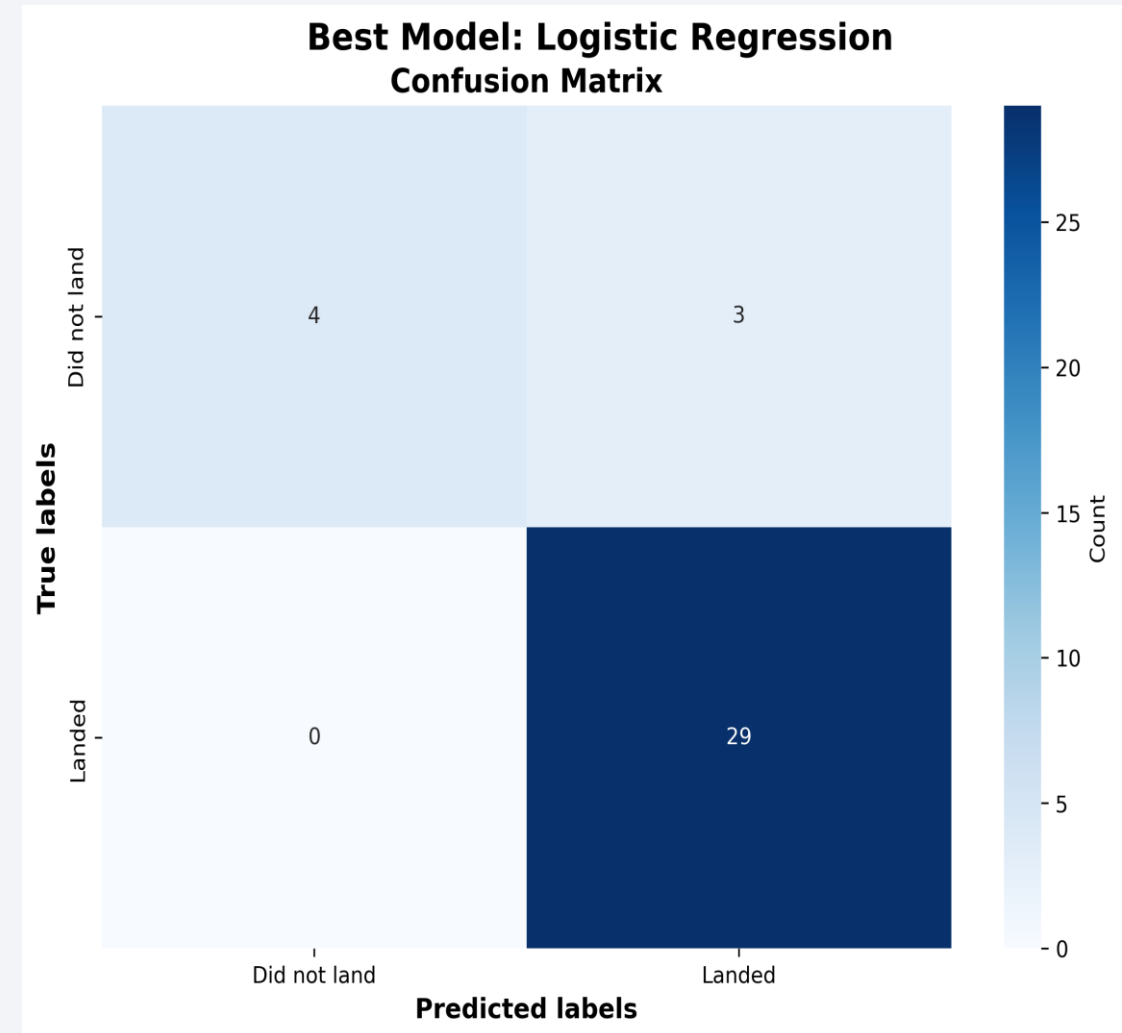


Confusion Matrix

Show the confusion matrix of the best performing model with an explanation

Based on the confusion matrix, Logistic Regression is the best model because:

- It never misses a successful landing (0 false negatives).
- It accurately classifies most launches, with only 3 minor errors.
- It achieves a high overall accuracy (91.7%) and maintains a balanced prediction performance between “landed” and “did not land.”



Conclusions

- The Data Science methodology effectively identified factors influencing launch success.
- Logistic Regression emerged as the best predictive model, with high accuracy and zero missed successes.
- The Data Analysis supports data-driven decision-making for future SpaceX missions optimizing launch conditions, payload limits, and booster usage to maximize success rates.

Thank you!

